

# Chapter 13 Summary — Behavioral Analysis Pipelines on Real Hand Histories

Research on the possibilities for applying Artificial Intelligence in computer games

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## 13 Chapter 13 — Behavioral Analysis of Real Hand Histories

Chapters 2–12 studied games that could be solved or simulated, where every strategy was known and every result could be checked against an exact reference. This chapter turns to what real people do. It takes two million online poker hands and asks what can be learned about the players from the record alone: how to describe a player, how well their decisions can be predicted, whether the same person can be recognized again, how stable a playing style is, whether two accounts are helping each other, and whether an account is a bot. The chapter builds the pipeline that does this, from parsing the raw files to scoring detectors against ground truth, and it measures every step. **All numbers reported here were measured** on reproducible runs, with three seeds wherever training or sampling is involved; the result files are named in the report. Where a run contradicted what the plan predicted, the prediction is kept and reconciled with what happened.

**The data.** The plan assumes hand histories from Playtech, which the candidate does not have yet. The substitute is public. The PHH dataset<sup>1</sup> contains 21.6 million anonymized no-limit hold'em hands from

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<sup>1</sup>Kim, J. (2024). “Recording and Describing Poker Hands.” *IEEE Conference on Games (CoG 2024)*, 1–8. DOI 10.1109/CoG60054.2024.10645611; arXiv:2312.11753 — the PHH format. The dataset is [github.com/uoftcprg/phh-dataset](https://github.com/uoftcprg/phh-dataset)

July 2009, collected from six online rooms by the hand-history vendor HandHQ. Its iPoker part includes **2,032,655 hands at \$0.50/\$1 blinds**, and these are used here. iPoker was Playtech’s own network at that time: Playtech launched it in 2004, and its 2009 annual report describes it as “Playtech’s exclusive poker network”<sup>2</sup>. So this is the closest public sample of play on a Playtech network, but it is 2009 data, not Playtech’s current data, and the Playtech dataset remains the validation this chapter prepares for. The second source is the **10,000 released hands of Pluribus**<sup>3</sup>, each played by the program and five professionals. It is the one public sample in which a bot is known, which makes it the ground truth for bot detection.

**Where this sits in the thesis.** The chapter is the practical core of Contribution #1, the behavioral adaptation framework. It covers the data side of the framework: representing behavior (statistics, embeddings), classifying it (clusters, Bayesian types), predicting it (behavioral cloning) and tracking it (stability over time, online inference), all from raw logs. The adaptive agent itself is outside the scope of this chapter. Real-time opponent modeling with a conservative fallback already exists and works in two-player games, including small poker games<sup>45</sup>. N-player opponent modeling exists too, but without any safety criterion<sup>6</sup>. So what C1 still needs is the N-player, within-match, safety-coupled version. Real logs are where that version will have to work, and this chapter measures how much information they carry. It also covers the fair-play task of the individual plan, detecting collusion and bots. In the literature, collusion detection is scored offline, as a detection task<sup>78</sup>. No counter-example was found to the statement that none of these methods couples detection to what the detecting agent does next. This chapter also stays on the offline side of that line. It measures what a detector can and cannot see, which a coupled agent will need to know.

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(MIT licence; HandHQ subset of 21,605,687 no-limit hands, July 2009, six rooms, 5,996,345 of them on the iPoker Network).

<sup>2</sup>Playtech Limited (2010). *Annual Report and Accounts for the year ended 31 December 2009*, pp. 1, 36–37 — “2004: Launch of iPoker network”; “Playtech’s exclusive poker network, iPoker”; the network “is designed to protect licensees and players against collusion and fraud”.

<sup>3</sup>Brown, N. & Sandholm, T. (2019). “Superhuman AI for multiplayer poker.” *Science* 365(6456), 885–890. DOI 10.1126/science.aay2400 — 10,000 hands against five of 13 professionals, 100 big blinds per hand; on style: limping is “suboptimal for any player except the ‘small blind’ player”, and Pluribus donk-bets “far more often than professional humans do”.

<sup>4</sup>Fu, H., Tian, Y., Yu, H., Liu, W., Wu, S., Xiong, J., Wen, Y., Li, K., Xing, J., Fu, Q. & Yang, W. (2022). “Greedy when Sure and Conservative when Uncertain about the Opponents.” *ICML*, PMLR 162, 6829–6848.

<sup>5</sup>Caen, A., Winands, M. H. M. & Soemers, D. J. N. J. (2026). “StratFormer: Adaptive Opponent Modeling and Exploitation in Imperfect-Information Games.” *Computers and Games 2026* (accepted); arXiv:2604.25796. See also Murgoci, V., Spaan, M. & Oren, Y. (2026). “AlphaExploit: Going beyond the Nash equilibrium in poker by learning to exploit suboptimal play.” arXiv:2605.09150 — both two-player.

<sup>6</sup>Ganzfried, S., Wang, K. A. & Chiswick, M. (2024). “Opponent Modeling in Multiplayer Imperfect-Information Games.” *Proc. 6th Int. Conf. Distributed Artificial Intelligence (DAI ’24)*, 39–45. DOI 10.1145/3719545.3721108.

<sup>7</sup>Mazrooei, P., Archibald, C. & Bowling, M. (2013). “Automating Collusion Detection in Sequential Games.” *AAAI* 27(1), 675–682. DOI 10.1609/aaai.v27i1.8674 — three-player limit hold’em, synthetic colluders. See also Greige, L. et al. (2022). “Collusion Detection in Team-Based Multiplayer Games.” arXiv:2203.05121 (social-graph features on real data, not poker).

<sup>8</sup>Bonjour, T., Aggarwal, V. & Bhargava, B. (2022). “Information Theoretic Approach to Detect Collusion in Multi-Agent Games.” *UAI*, PMLR 180, 223–232.

## 13.1 From solving games to reading logs

Picture a casino security room in which the cameras are off and only the dealer’s logbook is left. Each line says something like “seat 3 raised to 4, seat 5 folded”. A floor manager who reads enough pages learns each regular’s habits and notices two players who never raise each other. They also notice the account that plays three tables all day with the same rhythm. The pipeline in this chapter is that floor manager written down, with one addition the floor manager rarely has: a way to check how often each suspicion is right.

The logbook has limits. In the iPoker records, stacks are written as infinite, so an all-in is invisible. The winnings field is zero in every hand, so results have to be recomputed from the actions. Table names are identical, and hole cards are known only when shown, which leaves 86 % of hands with no known cards. Every statistic in this chapter is therefore defined on what the record contains, and the chapter says so where it matters. This is also the reason the chapter starts with a parser rather than a model. A mistake in parsing does not fail loudly; it quietly corrupts every number downstream.

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## 13.2 The data and the parser

The parser replays each hand with its own no-limit engine. The engine enforces turn order, minimum raises, stack limits and board counts, and it computes side pots and payoffs. A second, independent validator is pokerkit, a rules engine by the author of the PHH format<sup>9</sup>. The parsed hands become four data products, and each analysis reads one of them (the pipeline figure below).

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<sup>9</sup>Kim, J. (2025). “PokerKit: A Comprehensive Python Library for Fine-Grained Multivariant Poker Game Simulations.” *IEEE Transactions on Games* 17(1), 32–39. DOI 10.1109/TG.2023.3325637.

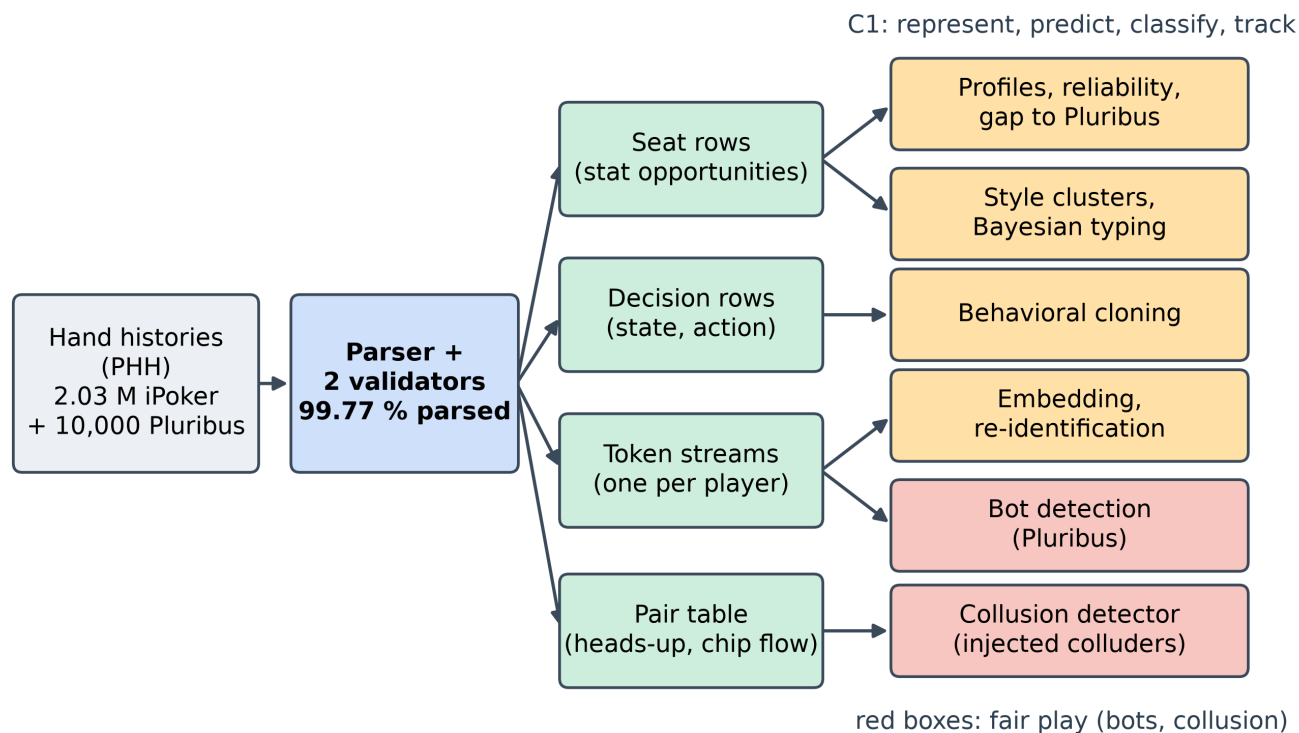


Figure 1: The chapter’s pipeline: raw PHH hand histories (2.03 million iPoker hands and the 10,000 Pluribus hands) are replayed by the parser and checked by two validators; the parsed hands become seat rows, decision rows, token streams and a pair table, each feeding one or two of the analyses. Red boxes are the fair-play analyses.

Three conventions of the format came to light only because two independent replays disagreed. In a heads-up hand the blinds are listed in reverse, so the big blind is player 1. A negative blind is not a refund; it is a *post*, the amount a newly seated player pays to join at once. The first version treated it as negative money, so a player who posted and folded appeared to win a big blind. The third convention concerns stacks: without them, an all-in shows up as a street dealt with no betting on it. The parser accepts exactly that pattern and marks the hand, which affects 4.3 % of hands.

With these handled, the parser replays 99.77 % of the iPoker hands and every Pluribus hand. The 0.23 % it rejects are genuinely incomplete, for example hands in which the big blind never acts. pokerkit accepts those, because an unfinished hand is not an illegal one. The two validators were then tested by injecting one error at a time into clean hands. The chapter’s validator caught all ten error types in every case. pokerkit caught illegal actions in every case, but it only warns on a duplicated card and does not flag a truncated hand. The two replays also agree on the payoffs of every comparable Pluribus hand and of 1,884 of 1,914 iPoker hands. The 30 differences are all showdowns where the cards were dealt face up while the show line reads “unknown”: there pokerkit splits the pot, and the parser evaluates the known cards. Used together, the two validators check both that each action is legal and that the hand as a whole is complete and consistent.

### 13.3 Describing a player: statistics and how many hands they need

Poker players describe each other with a handful of tracker statistics: how often a player puts money in voluntarily before the flop (VPIP), how often they raise before the flop (PFR), how often they re-raise a single raise (3-bet), how often the pre-flop raiser bets the flop (c-bet), how often a player who sees the flop goes to showdown (WTSD). Each of these is a count over its own *opportunities*. A 3-bet can only happen when a player faces exactly one raise, so the 3-bet rate is divided by those situations, not by all hands. Getting the opportunities right is where two trackers usually disagree, and it decides how many hands a statistic needs.

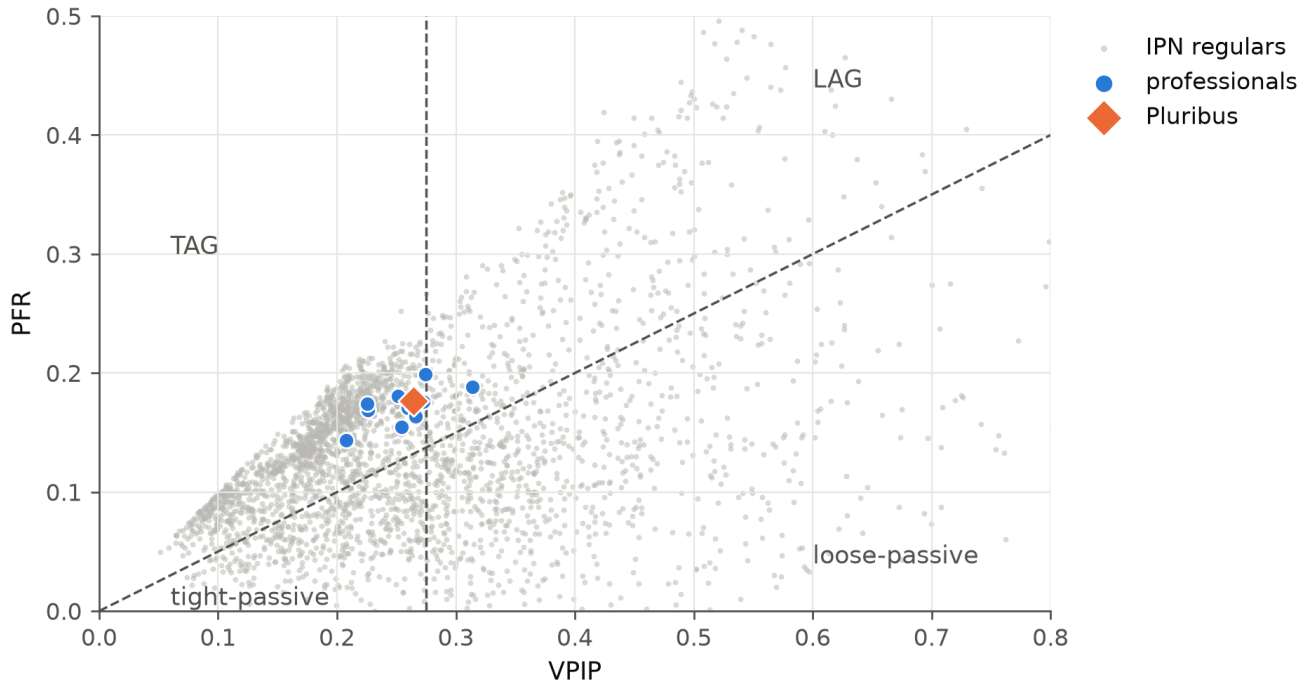


Figure 2: The classic VPIP  $\times$  PFR map for the 2,906 iPoker regulars (grey), with the 13 professionals of the Pluribus games (blue) and Pluribus itself (orange). Dashed lines: loose at VPIP  $\geq 27.5$  % (midway between the plan's tight 20 % and loose 35 %), aggressive when PFR  $\geq$  VPIP / 2.

The 2,906 players with at least 500 hands (the plan's threshold) hold 84.6 % of all seats. On the classic map, 1,323 are tight-aggressive, 730 loose-passive, 462 tight-passive and 391 loose-aggressive. The professionals and Pluribus sit together in a small patch at the loose edge of the tight-aggressive region.

How many hands does each number need? The honest way to answer is to measure it. The test takes two disjoint random samples of  $m$  hands from every player who has at least  $2m$ , computes a statistic on both, and correlates the two across players. A statistic that describes the player gives the same value on both samples; one that mostly describes the cards does not.

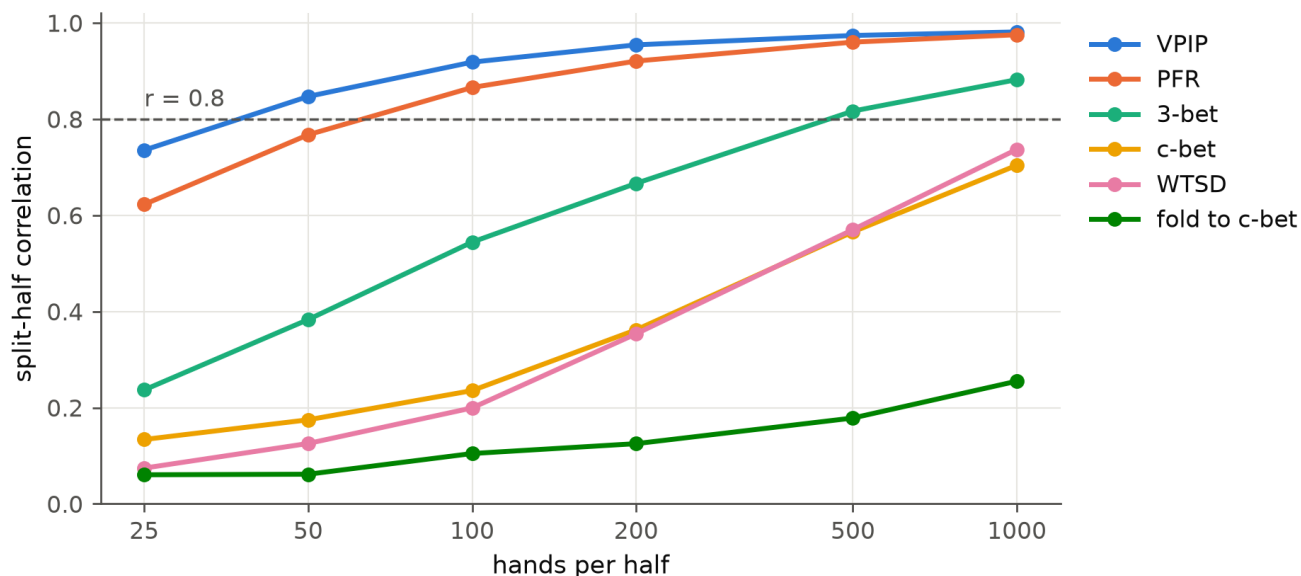


Figure 3: Split-half correlation of six statistics across the iPoker regulars as a function of the number of hands in each half (3 seeds; standard errors are smaller than the markers). VPIP and PFR pass  $r = 0.8$  below 100 hands, 3-bet near 500; the post-flop statistics stay below 0.8 at 1,000 hands.

The answers differ by an order of magnitude. VPIP reaches a correlation of 0.92 with 100 hands per half, and PFR 0.87. 3-bet needs about 500 hands to reach 0.82. The post-flop statistics are much slower: c-bet and showdown frequency are at 0.57 after 500 hands and about 0.7 after 1,000, and fold-to-c-bet is at 0.25 after 1,000. The plan’s rule of thumb, 500 hands for stable statistics, is right for the pre-flop numbers and wrong for the post-flop ones. The chapter therefore keeps 500 hands as the inclusion threshold and adds a minimum number of *opportunities* per statistic. Win-at-showdown is dropped altogether, because cards are known for only a median 41 % of a regular’s showdowns.

### 13.4 How far real players are from a strong reference

The plan asks for the gap between real play and equilibrium play, the “exploitation opportunity” that Contribution #2 would act on. For six-player no-limit hold’em no computed equilibrium is available. The defensible reference is the strongest six-player strategy with public hands: Pluribus’s own frequencies in its 10,000 six-handed hands. Pluribus is not an equilibrium, and a frequency gap is not an expected value. It says where an exploiter would look, not how much it would win.

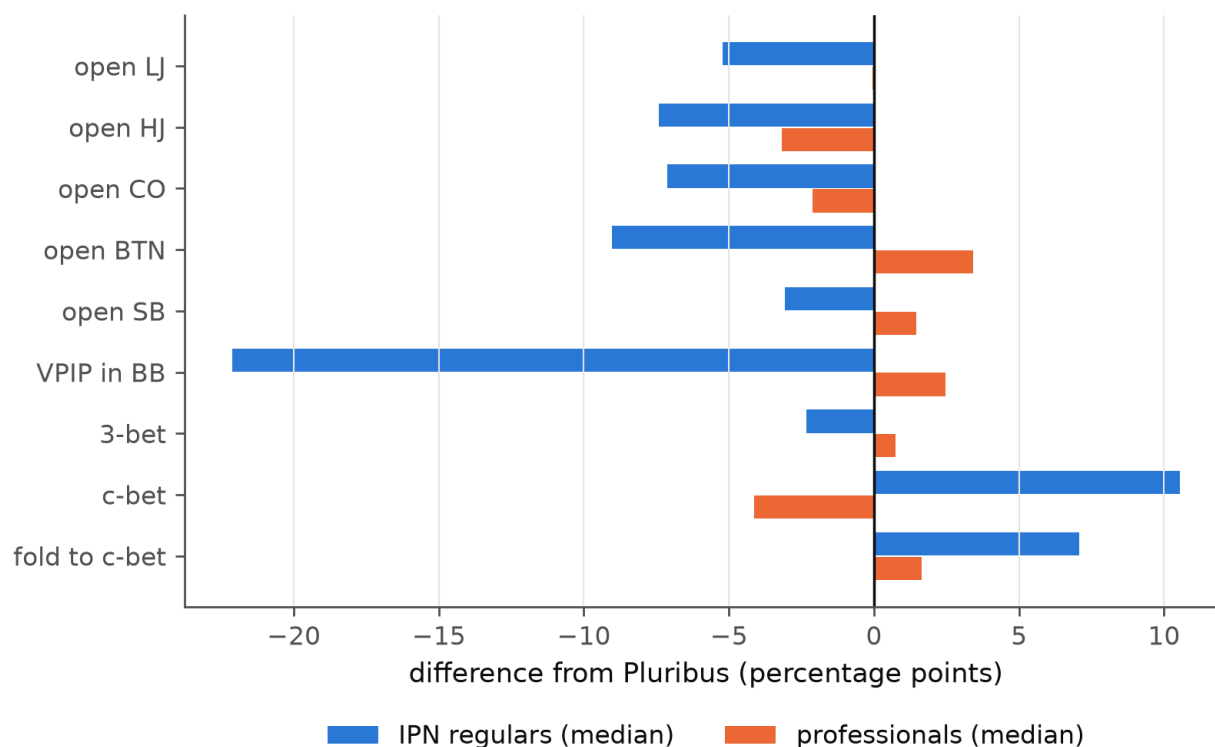


Figure 4: Median difference from Pluribus’s six-handed frequencies, for the 1,046 iPoker regulars with at least 500 six-handed hands and for the 13 professionals of the Pluribus games (percentage points). The regulars defend the big blind far less and c-bet far more; the professionals stay within a few points.

The two populations separate cleanly. Averaged over eleven pre-flop and flop frequencies, the median iPoker regular differs from Pluribus by 9.3 percentage points; the median professional by 3.9. Only 0.19 % of the 1,046 regulars with enough six-handed hands are closer to Pluribus than the median professional. The direction is consistent. The 2009 regulars defend the big blind far less: their median VPIP in the big blind is 22 points below Pluribus’s. They open fewer hands from every position, by 3 to 9 points, and 3-bet slightly less. After the flop they c-bet 10.5 points more often, and they fold to c-bets 7.1 points more often. The two flop gaps pull in opposite directions: these players bet the flop too often and give up on it too often. In the language of Chapter 8, these are the frequencies a safe exploiter would deviate against. Turning them into a value needs a model of the game that a frequency table cannot supply.

### 13.5 Predicting decisions: behavioral cloning and the baseline trap

Behavioral cloning is the simplest model of behavior: predict the action a player took from the situation they faced. The situation here is the public state — street, position, table size, players left and still to act, pot, price, raises so far, who was the aggressor, board texture. Hole cards are left out on the iPoker data, because no card is known in 86 % of hands and, where known, selected by having been shown. The actions are six classes: fold, check, call, and a small, medium or large bet or raise relative to the pot. The

model is trained on days 1–18 of the month and tested on days 19–25, so the test is the next week of the same people.

The plan set the target at 55 % accuracy, “with a random baseline of 14 %”. The model reaches 72.0 %, but that number says little. More than half of all decisions are folds, and a model that always predicts the most common legal action scores 70.2 %. The informative quantity is the gain over that baseline. A frequency table over the coarse situation adds 0.4 points, the neural network 1.4 points, and giving the network the acting player’s style vector — twelve statistics computed from the training days only — adds another 0.4 points. The same style vector lowers the log-loss clearly, from 0.800 to 0.751 nats (3 seeds, standard error 0.0002). This is the quantitative form of a C1 claim: representing *who* acts improves the prediction of what they will do, and it improves the calibration of the prediction more than its top-choice accuracy.

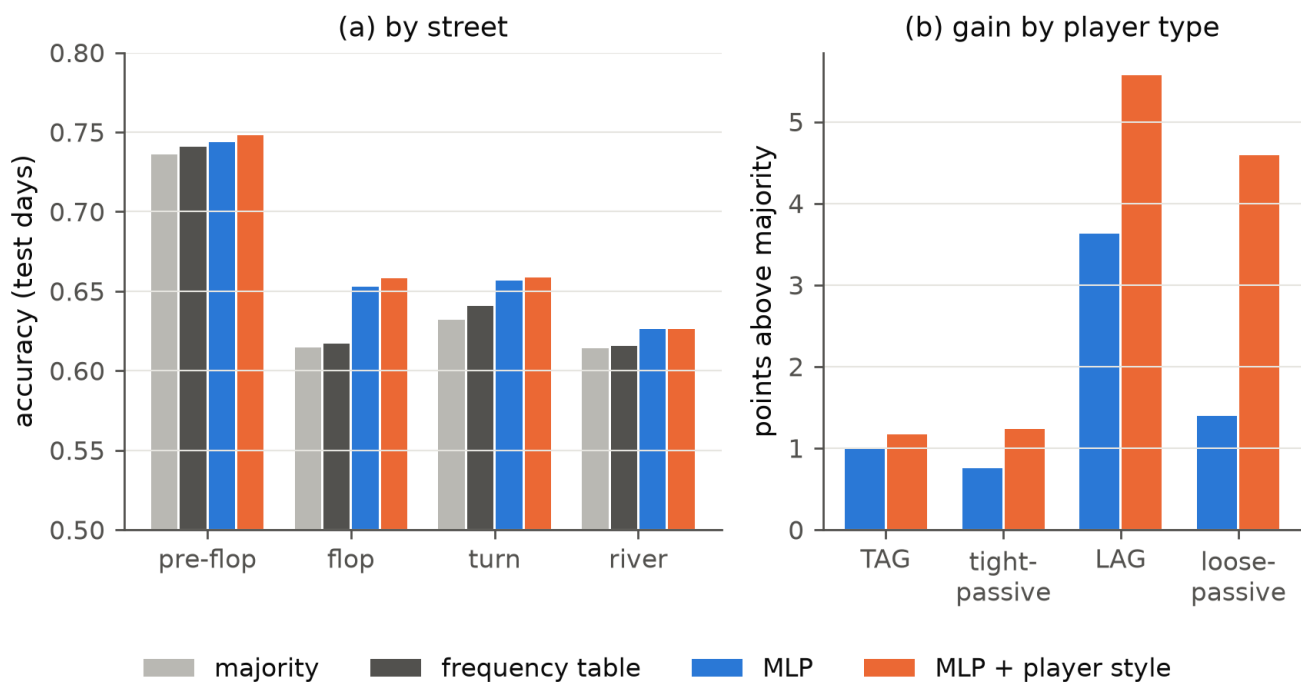


Figure 5: Behavioral cloning on the iPoker test days: (a) accuracy by street for the majority baseline, the frequency table, the MLP and the MLP with the player’s style vector (3-seed means); (b) accuracy above the majority baseline by player type. Knowing the player helps most for the loose players.

**Reconciliation (kept prediction → what actually happened).** The plan predicted that pre-flop decisions would be the most predictable, the button the most predictable position, and tight-aggressive players more predictable than loose-aggressive ones, and those more than weak loose players. Pre-flop is indeed the most predictable street (0.748 against 0.626–0.659 post-flop), but the majority baseline shows the same order. The button is the *least* predictable position (0.639) and early position the most (0.813). Raw accuracy does order the types as predicted (tight-aggressive 0.786, loose-aggressive 0.650, loose-passive 0.642). Measured *above the baseline*, the order reverses: +1.2 points for tight-aggressive, +4.6 for loose-passive and

+5.6 for loose-aggressive. I suspected a leak first and checked that the style vector uses only training days and that the test week is disjoint; it does, and it is. The predictions confused “easy to predict” with “folds a lot”. The lesson survives in a sharper form: the style vector adds most for the players furthest from the population (+3.2 points for loose-passive players, +0.2 for tight-aggressive ones).

On the Pluribus table every hole card is known, and the same network reaches 81.8 % (84.5 % with the style vector, majority 69.7 %), with 92.6 % before the flop. That is the size of the part of behavior that the iPoker data hides. Kumar et al.<sup>10</sup> argue that on noisy data with sparse rewards, offline reinforcement learning can beat cloning. Poker logs are exactly that kind of data, so cloning here is used only as a description of behavior, not as a policy to play.

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### 13.6 A behavioral fingerprint: player2vec for poker

Twelve statistics are a hand-made summary. The player2vec approach<sup>11</sup> replaces them with a learned one. Each in-game event becomes a token, and a Transformer<sup>12</sup> is trained to predict masked tokens from their context, as in BERT<sup>13</sup>. A player’s vector is then pooled from the model’s outputs over their event stream. The original paper applied this to a casual mobile game and read its clusters qualitatively; it did not test what the embedding captures.

For poker, each hand a player sits in becomes a short sentence. It opens with one token for position and table size, continues with one token per decision (street, whether a bet is faced, whether the pot is multiway, the action), and ends with a token for how the hand ended. A 4-layer encoder is trained on 256-token windows of each player’s stream from days 1–18 only. A first run with 4 epochs ended in different places for the three seeds; with 15 epochs all three pass the same sharp drop in loss, when the model learns the stream’s deterministic structure (positions rotate from hand to hand; the end token reveals a fold). The analyses use the 15-epoch models.

What should a player embedding capture? At least the player. That gives a test that needs no labels. Each player is embedded twice, once from N hands of the training days and once from N different hands of the last week, which the model never saw. Then, for every player’s last-week vector, all players are ranked by similarity. The question is whether the player’s own first vector comes top. The statistics vector gets the same test.

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<sup>10</sup>Kumar, A., Hong, J., Singh, A. & Levine, S. (2022). “When Should We Prefer Offline Reinforcement Learning Over Behavioral Cloning?” *ICLR 2022*; arXiv:2204.05618.

<sup>11</sup>Wang, T., Honari-Jahromi, M., Katsarou, S., Mikheeva, O., Panagiotakopoulos, T., Asadi, S. & Smirnov, O. (2024). “player2vec: A Language Modeling Approach to Understand Player Behavior in Games.” arXiv:2404.04234 (preprint).

<sup>12</sup>Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł. & Polosukhin, I. (2017). “Attention Is All You Need.” *NeurIPS 30*; arXiv:1706.03762.

<sup>13</sup>Devlin, J., Chang, M.-W., Lee, K. & Toutanova, K. (2019). “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.” *NAACL-HLT*, 4171–4186. DOI 10.18653/v1/N19-1423.

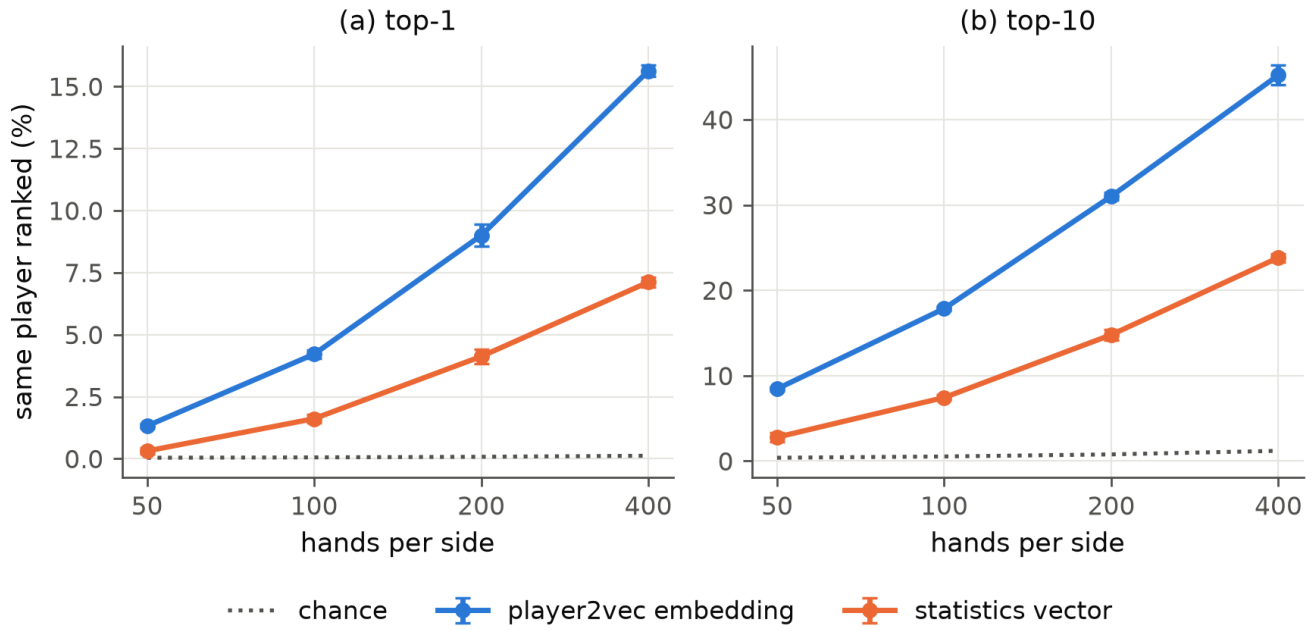


Figure 6: Re-identification of the same player on unseen days: share of players whose own later hands rank first (a) or in the top ten (b) among all candidates, for the player2vec embedding and the statistics vector, against the number of hands on each side (3 seeds,  $\pm 1$  SE). Chance is the dotted line.

The embedding recognizes a player at least twice as often as the statistics at every sample size. With 400 hands on each side it ranks the right player first among 834 for 15.6 % of them ( $\pm 0.2$ , 3 seeds) and in the top ten for 45.2 %, against 7.1 % and 23.8 % for the statistics; chance is 0.12 %. With 100 hands the figures are 4.2 % against 1.6 %. Averaging the outputs works much better than the max pooling of the original paper (4.5 % at 400 hands). Re-identification is also the multi-accounting question, whether one person can be recognized across accounts. The answer here is “sometimes, from a few hundred hands”: enough to rank candidates for a human to review, and far from proof.

The embedding also carries the classic axes, although it never sees a VPIP number. A fifteen-nearest-neighbor classifier recovers a regular’s VPIP  $\times$  PFR quadrant from the embedding with 90 % accuracy, slightly more than from the statistics vector that contains VPIP and PFR (88 %). Cosine similarity is 0.87 within a quadrant and 0.75 between quadrants, so the plan’s check “intra-cluster similarity greater than inter-cluster similarity” holds.

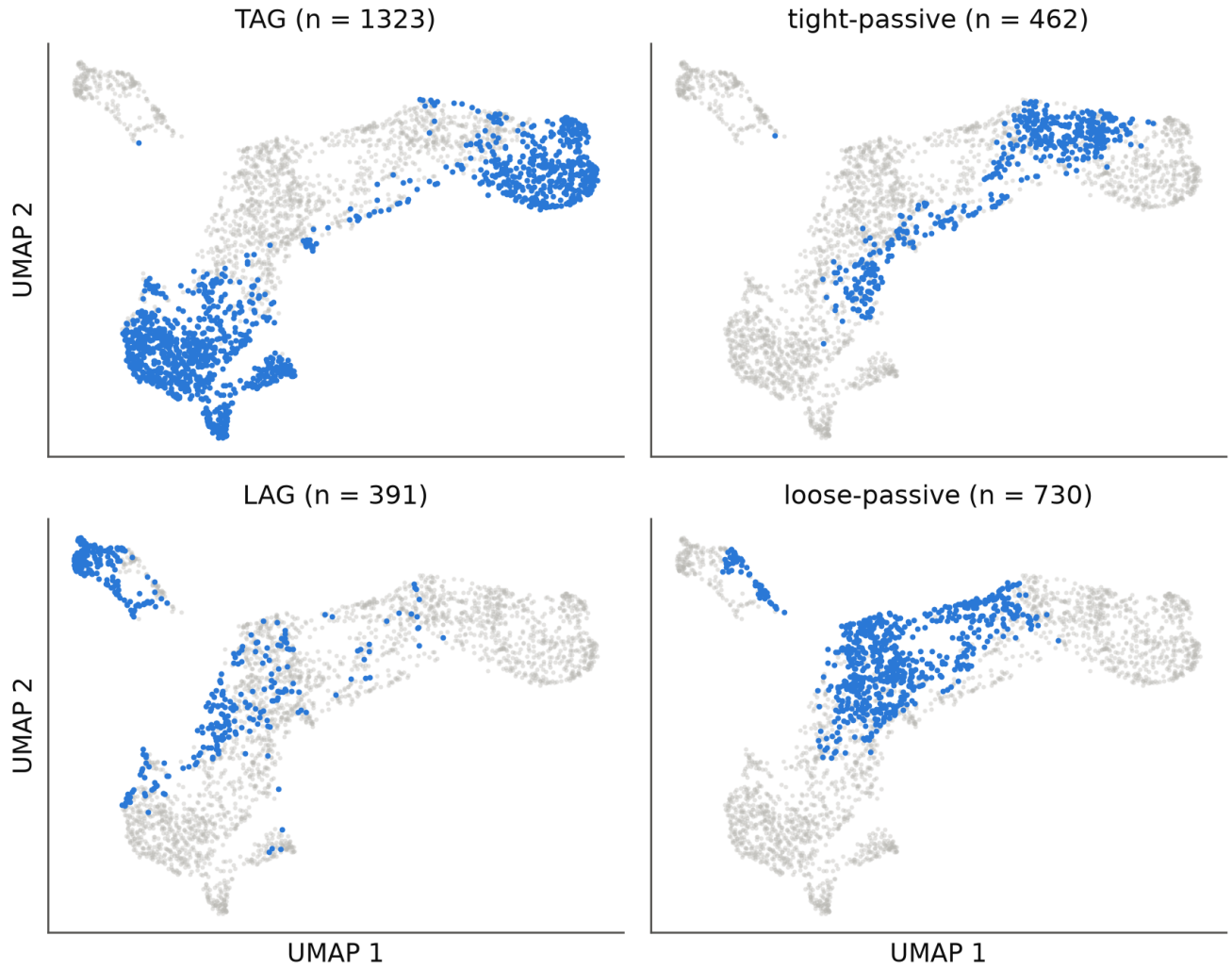


Figure 7: UMAP projection of the player2vec embeddings of the 2,906 iPoker regulars (seed 0), one panel per VPFR  $\times$  PFR quadrant, highlighted in blue over all players in grey. The quadrants occupy different regions of the map, with the loose-aggressive players partly in a separate island.

### 13.7 Types, stability and online inference

The plan expected four k-means clusters matching the four archetypes: tight-aggressive, loose-aggressive, tight-passive (“nit”) and loose-passive (“fish”). The data does not have four natural clusters. The silhouette<sup>14</sup>, a measure of how well separated clusters are, is 0.16 for two or three clusters and 0.13 for four, low values throughout. k-means with four clusters is stable across seeds (adjusted Rand index<sup>15</sup> 0.94 and 0.97), but what it finds is two tight-aggressive groups, one loose-aggressive and one loose-passive group,

<sup>14</sup>Rousseeuw, P. J. (1987). “Silhouettes: A graphical aid to the interpretation and validation of cluster analysis.” *Journal of Computational and Applied Mathematics* 20, 53–65. DOI 10.1016/0377-0427(87)90125-7.

<sup>15</sup>Hubert, L. & Arabie, P. (1985). “Comparing partitions.” *Journal of Classification* 2(1), 193–218. DOI 10.1007/BF01908075.

with no tight-passive cluster. Playing styles form a continuum with a dense tight-aggressive core, and “four archetypes” is a convention for describing it.

Do players stay in their cluster? The plan’s test splits each regular’s hands into the first and second half of the month and asks whether both halves land in the same cluster; the target is more than 70 %. The measured figure is 73.6 % ( $\pm 0.2$ , 3 seeds), so the target is met. A control shows how little that says: splitting the same hands *randomly*, which removes any change over time, gives 78.2 %. Most switching is therefore sampling noise at cluster borders, and genuine change over the month accounts for about five points. In the embedding space the stability is higher, 90.7 %, though its first half overlaps the model’s training days.

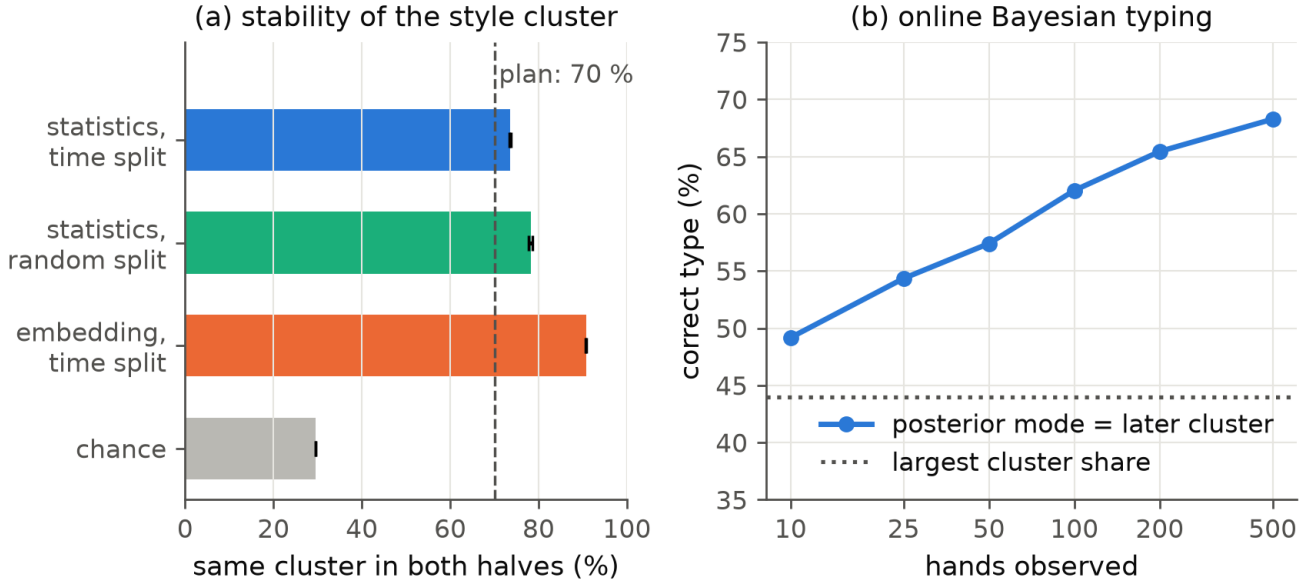


Figure 8: (a) Share of regulars in the same style cluster in both halves of their hands: chronological halves (statistics and embedding) and random halves (statistics), with chance; (b) online Bayesian typing: share of players whose posterior mode after  $n$  hands equals the cluster of their later hands, with the largest-cluster share for comparison.

Clusters are still useful as the *prior* of an online model, which closes a loop the plan left open: data  $\rightarrow$  clusters  $\rightarrow$  prior  $\rightarrow$  Bayesian update. This is the Bayesian opponent modeling of Chapter 7 and of *Bayes’ Bluff*<sup>16</sup>, applied to real players. The four clusters become four types, each with its own rates for per-hand events (voluntary entry, raise, limp; 3-bet, c-bet, showdown and steal per opportunity). The posterior over a player’s type is then updated hand by hand. The truth is the cluster of the same player’s *later* hands, so the test does not reward memorizing the data it predicts. Over the 1,702 players with at least 1,000 hands, the posterior’s best guess is right for 49 % after 10 hands, 62 % after 100 and 68 % after 500; always guessing the largest cluster scores 44 %. Two thirds of the players reach a posterior of at least 0.9 on their later type within 500 hands, after a median 85 hands. The ceiling near 70 % is the temporal

<sup>16</sup>Southey, F., Bowling, M., Larson, B., Piccione, C., Burch, N., Billings, D. & Rayner, C. (2005). “Bayes’ Bluff: Opponent Modelling in Poker.” *UAI 2005*, 550–558; arXiv:1207.1411.

stability above: a model cannot predict a later cluster more reliably than players stay in it.

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### 13.8 Collusion: detection on injected colluders in real sessions

Collusion in online poker takes three classic forms: colluders share information about their cards, they *soft-play* each other by not betting against a partner, or they *chip-dump* by losing on purpose to move money. No labelled collusion data is public. The main poker detectors were validated on synthetic colluders: collusion tables on three-player limit hold'em, whose authors state they knew of no suitable human dataset with known colluders<sup>7</sup>, and mutual information between agents' actions on Leduc hold'em<sup>8</sup>. Work on synthetic test benches argues that synthetic data should come first and be verified on real cases later<sup>17</sup>. (Two of the plan's reading-list papers, attributed to DeLong & Bhatt and to Yan & Browne, could not be found and are not cited.)

This chapter keeps the ground truth synthetic but puts the colluders into *real* sessions. For each seed, 40 pairs of real players are chosen who already share at least 300 hands and 30 heads-up confrontations, so that sitting together cannot give them away. In a share  $q$  of the hands where the two are left heads-up against each other, the rest of the hand is rewritten. For soft play, both players check or call to the end. For chip dumping, one bets the pot, the other raises to three times that, and the first folds. Every rewritten hand was accepted by both the parser and pokerkit, up to 3,306 hands per setting with no rejections, so the injected hands are legal poker, not noise.

The detector carries over Chapter 11's help/harm idea: compare what a player does *to the partner* with what they do to everyone else. Harm withheld is soft play: a player's aggression against the partner, standardized against the same player's aggression when heads-up against anyone else. Help given is dumping: the chip flow between the two in their heads-up hands, standardized by its own spread. The pair's score is the larger of the two. Chapter 11's score with raw event counts (a fold to the partner counted as help, a bet or raise at the partner as harm) was kept as a baseline.

The first run immediately flagged pairs with soft-play scores above 100. All of them came from heads-up *tables*, where the two players meet in every hand. The baseline had compared play at a heads-up table with being left heads-up at a six-player table, which is a different game. The headline run compares like with like and uses multiway tables only. This lowered the null's 99.9th percentile from 5.88 to 4.12.

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<sup>17</sup>Smed, J., Knuutila, T. & Hakonen, H. (2007). "Towards Swift and Accurate Collusion Detection." *GAME-ON 2007*, 103–107.

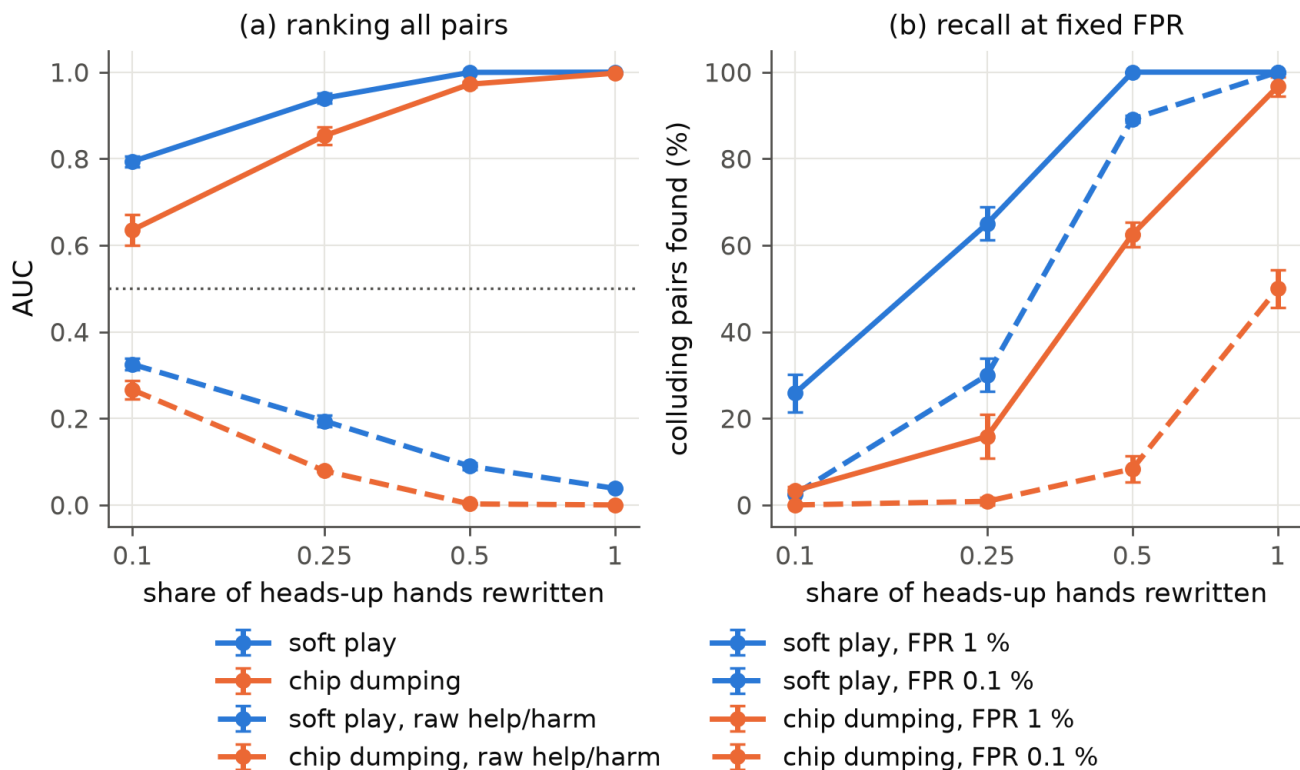


Figure 9: Collusion injected into real iPoker sessions (multiway tables, 3 seeds,  $\pm 1$  SE): (a) AUC of the union detector and of Chapter 11’s raw help/harm score against the share of heads-up hands rewritten; (b) share of colluding pairs found at a false-positive rate of 1 % and 0.1 % among 2,802 real pairs.

Against the 2,802 real pairs of the same stratum, which are presumed clean, soft play is the easier pattern. If a quarter of the confrontations are soft, the detector ranks the colluding pair above a random real pair 94 % of the time (AUC  $0.940 \pm 0.011$ ), and finds 65 % of the pairs at a 1 % false-positive rate. From half the confrontations upward it finds all of them. Dumping needs more: 62 % of pairs are found at 1 % false positives when half the confrontations are dumps, and 97 % when all are. Light collusion, one confrontation in ten, is mostly invisible (26 % and 3 % found). Detection also grows with the sample. At the null’s 99th percentile, soft play at  $q = 0.25$  is found for 48 % of the pairs with at most 39 heads-up confrontations, and for 83 % of the pairs with more.

**Reconciliation (kept prediction  $\rightarrow$  what actually happened).** The plan set the target at “detects 100 % of injected synthetic collusion at the chosen threshold, false positive rate on known-clean data  $< 5$  %”. Measured: met for soft play once half the confrontations are soft (false-positive rate 0.1 % at 90 % recall), and for dumping only when every confrontation is a dump (0.4 %). Chapter 11’s help/harm score does not transfer with raw counts: its AUC is 0.04–0.33 in every setting, worse than chance. I suspected an inverted sign first; the code is right, and the mechanism is in the data. Soft play removes the partner’s folds as well as the bets, so help falls with harm, and dumping *adds* bets, which counts as harm. The idea behind the detector survives; the raw counts do not. It works once each player is compared

with their own behavior, and once like is compared with like.

The highest scores among real pairs, 7.29, 4.71 and 4.16, lie above the null’s 99.9th percentile. They are candidates for a human to review, not findings; this is 2009 data with no labels, and one pair in a thousand lies above that percentile by construction.

### 13.9 Can the pipeline find a bot?

A bot in a human game is the other fair-play concern. The Pluribus hands allow a direct test: cut every player’s hands into blocks of 250, give the pipeline no labels, and see whether its anomaly scores single out Pluribus’s blocks. The Math Flag of the plan predicted that a bot would be *unusually consistent*, so its masked-token loss would be low.

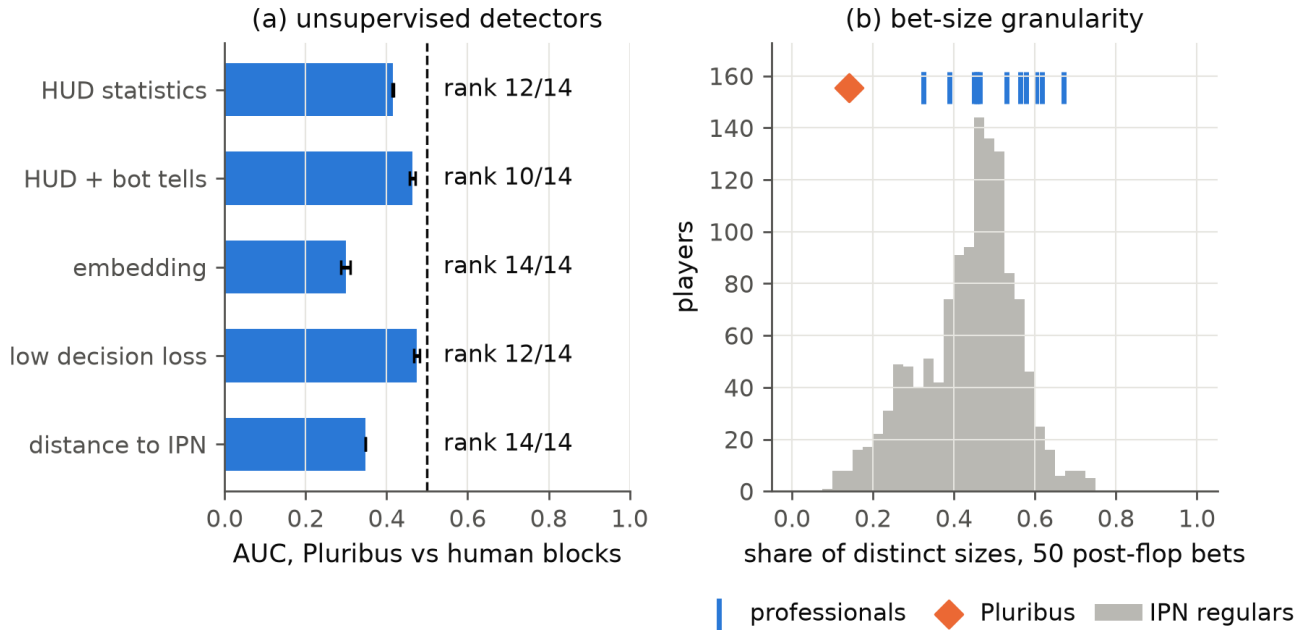


Figure 10: (a) Five unsupervised detectors on 250-hand blocks of the Pluribus games: AUC of Pluribus’s blocks against the professionals’ (3 seeds) and Pluribus’s rank among 14 players by mean score; (b) share of distinct bet sizes among 50 random post-flop bets of each player: iPoker regulars (histogram), the 13 professionals (ticks) and Pluribus.

None of the five unsupervised detectors finds Pluribus. Isolation Forest<sup>18</sup> on the HUD statistics scores an AUC of 0.415, on the statistics plus the features the Pluribus paper points at 0.463, and on the embedding 0.298. Low decision loss scores 0.475, and distance from the iPoker population 0.347. Pluribus ranks 10th to 14th of 14 players. Every AUC is *below* one half: Pluribus is more typical of these games than the average professional. Its decision loss, 0.691, sits inside the human range (0.529–0.829). A bot that plays

<sup>18</sup>Liu, F. T., Ting, K. M. & Zhou, Z.-H. (2008). “Isolation Forest.” *IEEE International Conference on Data Mining (ICDM)*, 413–422. DOI 10.1109/ICDM.2008.17.

a fixed mixed strategy is predictable in its *distribution* of actions, not in its actions. Low loss would flag a deterministic script, not an equilibrium-style bot.

Looking with the label shows two things that do separate it. First, Pluribus bets from a small menu of sizes. Among 50 of its post-flop bets, only 14 % of the sizes are distinct, against 32–67 % for the professionals, and 16 of 1,275 iPoker regulars (1.3 %) are as low. Second, it randomizes given its cards. Before the flop, with position, situation and exact starting hand fixed, its choice of action has an entropy of 0.143 nats, the highest of the nine players with enough data (humans 0.059–0.107). An operator sees hole cards, so this signal is available to one. The paper’s remark that Pluribus donk-bets more often than professionals is also visible (2.1 % of opportunities against 0.9 %), but too small to detect with. The size menu and the randomization were found by looking at one bot with its label known. They are hypotheses for the Playtech data, not validated detectors, and a bot designer who knows them could remove both.

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### 13.10 Honest notes, limitations, and where this hands off

**What held up.** The pipeline is solid. Two independent replays agree on payoffs except for one known, explained class of hands; injected errors are caught every time; every injected collusion hand is legal poker. The two headline modeling results carry three seeds each, with standard errors far below the effects. Knowing the player improves prediction, especially its calibration, and a self-supervised embedding recognizes players on unseen days about twice as often as the tracker statistics.

**What did not, and why it matters.** Four of the plan’s expectations failed on real data. Accuracy targets set against a random baseline are meaningless when most decisions are folds. The four textbook archetypes are not natural clusters. Chapter 11’s raw help/harm score points the wrong way on poker. And a strong bot is not an outlier. Each failure was checked for a bug before it was accepted, and each has a mechanism the chapter can name.

**Trust.** The data limits are real. There are no stacks, no winnings and few cards in the iPoker records. The negatives of the collusion test are presumed clean, not known to be. The bot test has one bot. And the gap is measured against Pluribus’s frequencies, not an equilibrium.

**Limitations.** (1) July 2009 on iPoker is not Playtech today: players, stakes and software have changed, and every threshold must be re-measured on the Playtech data. (2) The collusion is synthetic and covers two scripted patterns. Information sharing, the most common form, is not modeled and would not show in these signals. (3) The optional Decision Transformer on real data was not run. The iPoker outcomes are incomplete, and for the Pluribus games there is no simulator to evaluate a return-conditioned policy. Only the luck-driven leakage that Paster et al.<sup>19</sup> warn about could have been measured.

**Connections to other chapters.** Backward: the Bayesian typing is Chapter 7’s opponent model with learned types. The gap to Pluribus gives Chapter 8’s safe exploitation something real to deviate against.

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<sup>19</sup>Paster, K., McIlraith, S. & Ba, J. (2022). “You Can’t Count on Luck: Why Decision Transformers and RvS Fail in Stochastic Environments.” *NeurIPS* 35, 38966–38979.

The collusion detector is Chapter 11’s help/harm detector, standardized, and the luck warning of Chapter 12 is why cloning is used here as description only. Forward: Chapter 14 turns to evaluation methodology (C3), where collusion has so far been measured apart from play; the detectors, baselines and injection harness of this chapter are inputs to it. The thesis’s adaptive agent would use the online typing and the collusion signals inside its decisions (C1, C2), which is the coupling this chapter does not attempt.

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### 13.11 Key takeaways for the thesis synthesis

- **Parse before you model.** Three data conventions (reversed heads-up blinds, posts as negative blinds, hidden all-ins) each corrupted results until two independent replays were compared; the final parser replays 99.77 % of two million real hands.
- **Every number needs its baseline and its sample size.** 72 % action accuracy is 1.8 points over “the most common action”; 73.6 % style stability is 4.6 points *under* a random split; VPIP is reliable after 100 hands and fold-to-c-bet not after 1,000.
- **Representing the player helps (C1).** The player’s style vector lowers the prediction loss from 0.800 to 0.751 nats. A self-supervised embedding recognizes a player on unseen days 15.6 % of the time among 834 (7.1 % for the statistics, 0.12 % by chance).
- **Styles are a continuum, types are still a prior.** There are no four natural clusters, but an online Bayesian model over four learned types reaches a confident type within a median 85 hands.
- **Collusion detection works where each player is compared with themselves.** Strong soft play and dumping are found at 1 % false positives; light collusion is not. Raw help/harm counts fail on poker.
- **A strong bot is not an outlier.** Unsupervised detectors rank Pluribus 10th–14th of 14. Its bet sizes and its randomization separate it, and these are hypotheses for the Playtech data.